

ABSTRACT

Multi-agent artificial intelligence systems increasingly need to learn collaboratively from decentralised experience without exposing raw, often privacy-sensitive, local data to a central authority or to one another. Federated learning offers exactly this capability, and its fusion with multi-agent reinforcement learning — known as Federated Multi-Agent Deep Reinforcement Learning (FedMARL) — has become an active research direction across wireless networks, cybersecurity, and aerial robotics. This seminar presents a comparative study of recent IEEE works that apply FedMARL to distinct real-world coordination problems.

Li et al., the base paper, formulate vehicle-to-vehicle communication as a multi-agent problem where each link is an agent trained via a dueling double deep Q-network, with local models periodically federated to overcome partial observability and speed up convergence. Chang et al. tackle a related spectrum-sharing problem using a fingerprint-based deep Q-network with a shared reward, enabling a fully distributed alternative to server-based averaging. Moudoud et al. extend the paradigm to cybersecurity, letting wireless sensor network nodes collaboratively detect zero-day threats without sharing raw traffic. Chen et al. apply it to aerial robotics, embedding a physical motion model in a federated RL loop so a UAV swarm can learn energy-efficient data-collection trajectories.

Comparing these systems across agent design, coordination structure, robustness to non-IID data, and real-world deployability reveals a consistent pattern: federating locally trained policies rather than raw data lets systems as different as connected vehicles, spectrum-sharing links, sensor networks, and UAV swarms reach faster, more robust joint behaviour than independent learning — at the cost of coordination overhead and, where a server is used, a single point of trust. The study concludes that hybrid FedMARL architectures, combining domain-specific agent design with trust-aware aggregation, are the most promising direction for scalable, privacy-preserving multi-agent AI.

Keywords — Multi-Agent Artificial Intelligence; Federated Learning; Federated Multi- Agent Deep Reinforcement Learning; Deep Q-Network; Vehicle-to-Vehicle Communi- cation; Dynamic Spectrum Access; Wireless Sensor Network Security; UAV Trajectory Planning.